

Double Pendulum and The Chaos Theory

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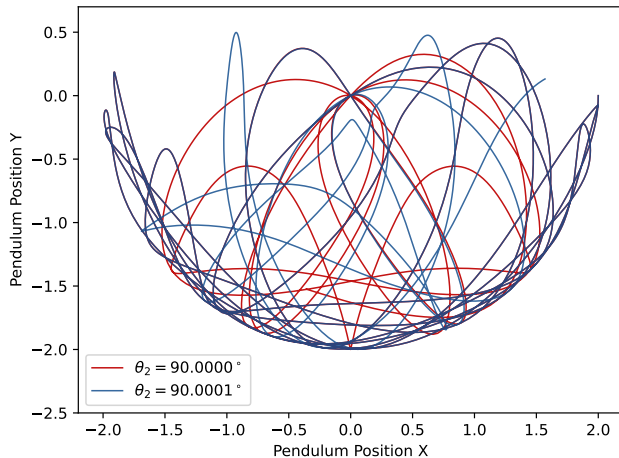
Related CALCAB Knowledge

Differential Equation: It is the core of simulation algorithm we applied in this project, the entire simulation was run based on a differential equation derived by *Lagrangian Mechanics*, so the differential equation we learned in unit 7 is the core of our project.

Limit and Approaching: In our project, the observation of the simulation accuracy as time step changes is also a core part. By speculating the result approaches a value as step decreases (becomes more accurate), we could choose a suitable time step for simulation that is both efficient and accurate.

Introduction To The Chaos Theory

Two Slightly Different Pendula Simulation Outcome



Uncertain Outcome: In a chaotic system described in the Chaos Theory, even a *negligible* disturbance or initial differences could cause an incredibly different result.

Determinable though Chaotic: Chaotic doesn't mean random and indeterminate, instead, chaotic systems could be predicted by calculations and modeling if one has incredibly precise data, though the result is input-sensitive.

Double Pendulum, Where Newtonian Physics Dies

Introduction To DE Calculation Approaches

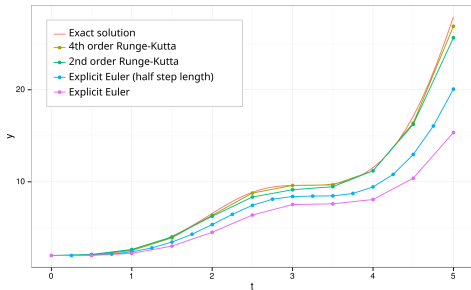


Figure: Comparison of the Runge-Kutta methods for the differential equation. Red line is the exact solution, 4th order Runge-Kutta is what we mainly use in this project, and is significantly more accurate. Figure from Wikicommon, CC BY-SA 3.0

Fourth-Order Runge-Kutta (RK4) Approach:

$$\text{Let } \frac{dy}{dx} = f(x, y),$$

$$y_{n+1} = y_n + \frac{\Delta x}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

Where,

$$k_1 = f(x_n, y_n)$$

$$k_2 = f\left(x_n + \frac{\Delta x}{2}, y_n + \frac{\Delta x}{2}k_1\right)$$

$$k_3 = f\left(x_n + \frac{\Delta x}{2}, y_n + \frac{\Delta x}{2}k_2\right)$$

$$k_4 = f(x_n + \Delta x, y_n + \Delta x k_3)$$

